

Wireless Gesture-Controlled Robot

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In this project we are going to control a robot wirelessly using hand gestures. This is an easy, user-friendly way to interact with robotic

systems and robots. An accelerometer is used to detect the tilting position of your hand, and a microcontroller gets different analogue values and generates command signals to control the robot. This concept can be imple-

mented in a robotic arm used for welding or handling hazardous materials, such as in nuclear plants. The author's prototype is shown in Fig. 1.

Circuit and working

The block diagram of the wireless gesture-controlled robot is shown in Fig. 2. The circuit diagram of the wireless gesture-controlled robot is shown in Fig. 3 and of the receiver section in Fig. 4.

ATmega328. ATmega328 is a single-chip microcontroller from Atmel and belongs to the mega AVR series. The Atmel 8-bit AVR RISC based microcontroller combines 32kB ISP flash memory with read-while-write capa-

bilities, 1kB EEPROM, 2kB SRAM, 23 general-purpose I/O lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, serial programmable USART, a byte-oriented 2-wire serial interface, SPI serial port, 10-bit A/D converter, programmable watch-dog timer with an internal oscillator and five software-selectable power-saving modes.

The device operates between 1.8 and 5.5 volts. It achieves throughputs approaching one MIPS per MHz. An alternative to ATmega328 is ATmega328P.

ADXL335. This is a complete three-axis acceleration measurement system. ADXL335 has a minimum measurement range of $\pm 3g$. It contains a poly-silicon-surface micro-machined sensor and signal-conditioning circuitry to implement open-loop acceleration measurement architecture. Output signals are analogue voltages that are proportional to acceleration. The accelerometer can measure the static acceleration of gravity in tilt-sensing applications as well as dynamic acceleration resulting from motion, shock or vibration.

The sensor is a poly-silicon-surface micro-machined structure built on top of a silicon wafer. Poly-silicon springs suspend the structure over the surface of the wafer and

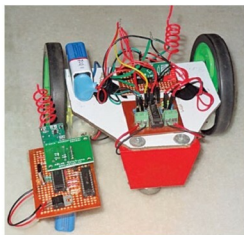


Fig. 1: Author's prototype

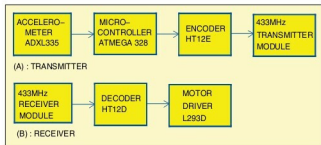


Fig. 2: Block diagram of the wireless gesture-controlled robot

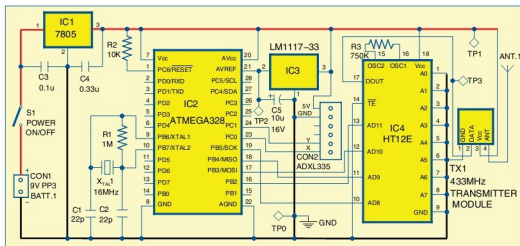


Fig. 3: Transmitter section of the wireless gesture-controlled robot

